

0. Reframed Objective and Design Philosophy

Problem: No robotic system achieves 100% anything in collapsed structures. DARPA SubT drones — far heavier and better-funded than nano-drones — routinely failed at doorways and were lost. Optimizing for zero loss forces conservatism that defeats the mission.

New objective:

Maximize the probability of locating live victims per unit time, under bounded and predictable drone attrition, such that every drone's information survives the loss of the drone itself.

This is the honest, achievable version of "risk-free": the system removes humans from the immediate danger zone during the initial search, and the *information* a drone gathers is guaranteed to reach the operator even when the drone does not return.

Seven governing principles:

1. **Victim detection is the spine, not an add-on.** The radar you already carry is dual-use: navigation *and* vital-signs detection.
2. **Drones are semi-expendable sensors.** Budget for attrition; make data outlive the drone.
3. **Infrastructure beats dead-reckoning.** Drop static anchors to bound drift, instead of trusting a flying mesh in degenerate tunnel geometry.
4. **Information must flow out continuously.** A dropped-node radio backbone carries life-critical alerts in real time; big maps wait.

5. **Heterogeneous, layered fleet.** Match capability (and cost) to role; don't put \$455 of compute on a beacon-carrier.
 6. **Robust classical coordination over fragile learned policies.** Auction-based frontier allocation, not end-to-end MARL.
 7. **Design around 2–4 minute sorties.** Endurance is the dominant constraint; the mission is waves, not a single sweep.
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1. Hardware: A Three-Tier Heterogeneous Fleet

Replacing the uniform 50× identical fleet. Heterogeneity cuts cost and matches capability to role.

Tier 0 — Base / Gateway (1 unit, at the entrance)

- Operator station + high-compute server (global pose-graph optimization, dense map reconstruction, victim-alert dashboard).
- High-gain antenna anchoring the radio backbone.
- Charging / battery-swap station for wave-based relaunch.
- Optionally a wheeled rover or heavier "mothership" multirotor that can ferry a magazine of relay beacons to the entrance and drop the first few nodes. (SubT winners used exactly this kind of ferry role.)

Tier 1 — Relay / Anchor Beacons (cheap, dumb, static — the backbone)

The single most important addition to your plan. Small battery-powered pucks combining **UWB anchor + low-rate radio repeater**, dropped or perched by the scouts as they advance.

- UWB anchor + low-rate radio repeater + **anchor-to-anchor UWB ranging** (pucks range to each other — cheap, and it stiffens the localization chain against drop-time position error).

- **Comms function:** multi-hop relay back to base (Section 6).
- Carried 1–2 at a time by scouts, or in bulk by the Tier-0 ferry near the entrance.
- These are expendable and *expected* to be left behind. They are the persistent layer that outlives any single drone's battery.

Why this is non-negotiable: every winning SubT team dropped battery-powered mesh nodes "like breadcrumbs... just before they passed out of communication range," because comms was lost "after the first bend in the tunnel." Your plan's flying-only mesh has the same failure mode they all hit and solved this same way.

Tier 2 — Nano-Scouts (Crazyflie 2.1 class, the penetrating layer)

Back-of-envelope: a Crazyflie-class platform has ~10–15 g usable payload, less after radar+UWB+IMU; a Tier-1 puck (UWB + repeater + hours-long battery) is plausibly 10–20 g. Scouts carrying '1–2 pucks' may not close — if not, the Tier-0 ferry/rover becomes load-bearing for the entire backbone and Wave 1 changes accordingly. Evaluate the Crazyflie 2.1 Brushless for the extra thrust margin.

Two sub-types, so you don't pay for GAP9 fifty times:

2a. Anchor/Mapper drones (~6 in a 50-unit fleet) — full GAP9Shield

- 24 GHz (consider 60 GHz) mmWave radar — RIO + obstacle avoidance + vital-signs modes.
- IMU (RIO/TIO core), UWB transceiver, barometer + downward ToF (altitude — see §2).
- **FLIR Lepton-class thermal microcamera (~0.9 g)** — sees through smoke, detects body heat; dual-use for thermal-inertial odometry *and* victim detection. (Thermal-inertial odometry has been shown to keep drift under ~2% over kilometer-scale GPS-denied flight on embedded hardware.)

- MEMS microphone array — passive acoustic victim cue (calls, tapping); near-zero power.
- Optional CO₂ sensor — confined victims raise local CO₂.
- GAP9Shield (~6 g, ~150 int-GOp/s, 32 MB L3) runs: radar point-cloud processing, the tiny place-recognition encoder, obstacle CNN, and the local EKF.

2b. Light Scouts (the remainder) — minimal/no GAP9

- Radar (obstacle + ego-velocity) + IMU + UWB + a thermal pixel array or single-zone thermal.
- Job: reactive obstacle avoidance, relay trajectory at low rate, **carry and drop one Tier-1 beacon**, act as extra ranging targets, and serve as static sensors after landing.
- Much cheaper; treat as the most expendable.

Endurance reality (state it plainly)

A Crazyflie is ~27–30 g with ~7 min flight time **bare**. Stacking radar + UWB + GAP9Shield realistically yields **2–4 minutes** of flight. The entire concept of operations (§7) is built around that number rather than wishing it away.

2. State Estimation: Bounding Drift Instead of Hoping It Stays Small

2.1 Odometry core — fuse two degraded-environment modalities

Run a tightly-coupled **Radar-Inertial + Thermal-Inertial Odometry** front-end into a fixed-lag smoother / EKF. Both modalities survive smoke; fusing them gives redundancy when either degrades (radar

gives no returns when static; thermal struggles with low scene contrast). Per-drone state:

$X_i = [p_i, v_i, q_i, b_a, b_g]^T$
position, velocity, orientation, accel bias, gyro bias

Crucially, estimate the IMU biases explicitly. The documented killer of radar-inertial odometry is acceleration-bias drift that "grows linearly with time... and may diverge arbitrarily." Online bias calibration is not optional.

- Note: thermal-inertial is a real-hardware-validation-only component — Gazebo thermal modeling of low-contrast tunnel scenes is weak, so it will not be sim-validated alongside the radar-inertial core.

2.2 Altitude is the weak axis — reinforce it

Radar-inertial elevation drift is the worst-observed dimension, especially with vertical motion. Do **not** trust radar/IMU alone for height. Fuse **barometer + downward ToF + anchor geometry** for the vertical channel.

2.3 Drift bounding via dropped UWB anchors (the key fix)

A squad strung along a corridor is nearly collinear → catastrophic geometric dilution of precision → lateral/vertical position barely observable from neighbors. Solution:

- As scouts advance, they **drop Tier-1 UWB anchors**
 - Dropped anchors do not have known positions — each is known only as well as the dropping drone's drifted odometry at drop time (a surveyor's traverse). Anchor positions must therefore be treated as **state variables to be refined** (multi-pose ranging, anchor-to-anchor ranging,

loop closures propagating back through the anchor graph). And because a dropped puck lands on the floor, anchors are roughly **coplanar** → **poor vertical DOP**, compounding the weak Z axis; the baro+ToF fusion in §2.2 is what carries vertical, not anchor geometry.

- This converts "hopeless pure flying-mesh SLAM" into "infrastructure-anchored localization," which is tractable.

The anchors define a single global frame for the whole mission — which is also what makes map-merge tractable later (§9).

2.4 Loop closure — the legitimate, scoped home for the learned model

A **tiny** quantized place-recognition encoder on GAP9 turns a radar+thermal keyframe into a compact descriptor (think Scan-Context-style for radar, learned). Matching descriptors across the pose graph creates loop-closure factors that snap the graph back into consistency. This is where your "JEPA-style" representation idea belongs — **radically shrunk to a sub-1-GOp encoder**, doing place recognition only, not world modeling.

3. Mapping: Hybrid Topo-Metric (Cleanly Separated Layers)

Your "ST-JEPA topological factor graph" fused three different things. Separate them:

Layer	What it is	Where it lives	Purpose
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Metric local	small rolling occupancy / ESDF-lite around the drone, from radar	on-drone	real-time obstacle avoidance
Topological global	sparse pose graph: nodes = anchors + keyframes; edges = traversable links with relative transforms	on-drone (sparse) + base	navigation, RTH, squad fracturing, merge
Learned signature	per-node place descriptor + compact scene signature	on-drone (tiny encoder)	loop closure; lightweight scene tagging
Dense reconstruction	full 3D render for the operator	base station only	human situational awareness

Rule: heavy optimization (global pose-graph relaxation, dense reconstruction) runs at base. The nano-drone keeps only the sparse topological graph it needs to navigate and return.

4. Coordination: Auction-Based Frontier Allocation (Replaces MARL)

4.1 Why

Learned multi-agent policies are exactly where "research that doesn't work in real life" lives — sim-to-real transfer in rubble is brutal and unverifiable. Market-based allocation is the battle-tested SubT-grade alternative: decentralized, tiny-compute, inspectable.

4.2 Exploration backbone — frontiers

Each agent continuously detects **frontiers** (boundaries between known and unknown space) from its local map.

4.3 Two coordination regimes (because the mesh *will* fragment)

Connected (within mesh/relay range): auctions.

- Frontiers are auctioned (sequential-single-item, or combinatorial when synergies matter).
- Bid = **expected information gain - travel cost - battery risk - comms-stretch penalty.**
- Highest bidder claims the frontier; claims broadcast to deconflict.
- **Squad fracturing** at a doorway/branch is simply an auction over that new frontier; the winning Anchor peels its squad off. (Keeps your elegant fracturing idea, grounded in a proven mechanism.)

Disconnected (left the network — the common case): response-threshold dispersion.

- No negotiation possible, so coordinate *implicitly*: pick frontiers that minimize overlap with last-known peer positions; spread out deterministically.
- This directly handles the SubT reality that "robots will leave the communication network range soon."

The hybrid matters because the literature is explicit: auctions win when information is good; threshold methods match them "at a fraction of the expense" when information is poor.

4.4 Keep Anchor/Shadow leader-election — but generalize it

Your 1:1:8 redundancy is essentially Raft, and it's good. Improvement: have **every** squad member buffer a compressed copy of the squad's sparse pose graph, so losing the Anchor degrades but never destroys

the squad's accumulated map. The Shadow promotes; a new Shadow is elected; the graph is intact.

5. Victim Detection: The Actual Mission (New, Central)

5.1 Multi-cue detection

Fuse four independent signals into a confidence score:

1. **Thermal hotspot** (body heat through smoke) — fast, in-flight, coarse.
2. **Radar vital-signs** (breathing/heartbeat) — slow, near-stationary, confirmatory; works through light rubble at 0.5–2.5 m.
3. **Acoustic** (calls, tapping) — passive, near-zero power, omnidirectional cue.
4. **CO₂** (optional) — confined-space corroboration.

Thermal/acoustic are *cueing* sensors (cheap, always-on); radar vital-signs is the *confirming* sensor (expensive in time, needs stillness).

5.2 The perch-and-stare behavior (essential, and new)

Radar vital-signs detection is swamped by UAV ego-motion (chest displacement is millimeter-scale). So:

IF (thermal or acoustic cue exceeds threshold):

- fly to the cue, then PERCH on stable rubble (or hold the steadiest possible hover)
- run 10–30 s radar vital-signs integration (clutter suppression + breathing/heartbeat extraction)
- IF vitals confirmed: emit Class-A victim alert (see §6) and mark node in the graph

→ resume exploration or hold as a beacon over the victim

Perching also conserves battery (rotors idle), partially mitigating the endurance problem.

5.3 The most important output

On a confirmed detection, emit a **tiny high-priority packet** — `{victim_id, position_in_anchor_frame, confidence, vitals_estimate, timestamp}` — that propagates back over the relay backbone to the operator **immediately**. Finding one breathing person in the first ten minutes is the entire point; this packet is the system's reason to exist.

6. Communications & Information Return

Fixes the "offload only at base" flaw. The Tier-1 dropped nodes are simultaneously the **radio backbone**.

6.1 Two information classes

- **Class A — instant, tiny:** victim alerts, drone-down events, frontier claims, anchor placements. Pushed in real time over the multi-hop backbone even at very low bandwidth.
- **Class B — deferred, large:** full sparse pose graphs and any dense data. Offloaded at **rendezvous points** or on return — your asynchronous-offload idea, preserved.

6.2 Rendezvous behavior

Drones that have pushed past the live network periodically **return to the nearest live relay node**, sync Class-B data, then push deeper.

(The NeBula/CoSTAR pattern: leave the mesh, come back to a rendezvous to exchange, repeat.)

6.3 Why this beats the original

The operator gets actionable victim locations within seconds of detection instead of waiting for a drone to fly all the way home — and a lost drone has *already* contributed its alerts before it died.

7. Endurance Strategy: Waves, Not a Sweep

Built around the honest 2–4 minute sortie.

- **Wave 1 — establish the spine.** Early scouts push in, dropping the UWB/relay backbone along the main artery and establishing the global frame + comms hops near the entrance.
 - **Wave 2+ — penetrate.** Later waves launch into a now-localized, now-connected corridor, reaching deeper because they don't have to re-establish infrastructure.
 - **Perch-to-consume / perch-to-listen.** A drone low on battery lands at a relay node and becomes a static sensor (vital-signs watch, acoustic listen) rather than burning its last charge flying.
 - **"Become infrastructure" over risky RTH.** A drone that's low but sitting on a useful vantage (e.g., over a detected victim, or at a comms-critical junction) should *stay and serve* rather than attempt a long return it may not survive.
 - **Battery swap / relaunch at Tier-0** for the reusable Anchor drones between waves.
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8. Return-to-Home & Attrition Management (Reframed)

Keep elastic breadcrumb RTH, but make the trigger a decision, not a reflex.

At the bingo-fuel threshold, the policy chooses among:

1. **RTH** along the inverted topological path (if margin is sufficient and the path is still valid).
2. **Become a relay/sensor** (if positioned usefully and margin is marginal).
3. **Perch-and-stare on a detected victim** (if a live cue is present — its data matters more than the airframe).

This is an expected-value choice, not an automatic retreat.

The honest answer to "100% recoverable": you cannot guarantee the drone comes back. You *can* guarantee — via the dropped-node backbone and the instant-alert channel — that the drone's information comes back. Budget for an explicit attrition rate per mission and design so it doesn't compromise the result.

9. Map Synthesis at Base (Fixing "Easily Stitches")

Merging maps that dead-reckoned apart is the loop-closure/map-merge problem — not trivial. But the anchor backbone makes it tractable:

- Every squad expresses its pose graph **relative to shared, globally-known UWB-anchor positions**. Those anchors are **common landmarks across all partial maps**.
- Base-station merge = **pose-graph optimization** with:
 - **anchor-coincidence constraints** (the same physical anchor seen by two squads must coincide), plus

- **overlap-based loop closures** (place-recognition descriptor matches across squads).
 - Output: one consistent global pose graph → dense reconstruction → operator render, with **victim markers already delivered live** now placed in context.
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10. Honest Capability Statement & Failure Modes

What this can plausibly do: establish a localized, comms-connected corridor into a smoke-filled GPS-denied structure; detect and confirm live victims via multi-cue + perch-and-stare; stream victim locations to the operator in near-real-time; and return a merged map.

Known hard risks and mitigations:

- **Doorway/clutter traversal in the dark** — the documented Achilles' heel of SubT drones. Mitigation: nano size helps fit gaps; fused radar+thermal obstacle avoidance; accept that some passages are impassable and have the topological map *record* "blocked" rather than retry indefinitely.
- **Radar altitude drift** — mitigated by baro + ToF + anchor geometry (§2.2), but remains the weakest axis; validate it explicitly.
- **Vital-signs false positives/negatives** — require multi-cue agreement and a minimum integration window before emitting a Class-A alert; report confidence, never certainty.
- **Backbone gaps** — if a drop spacing exceeds radio/UWB range, a localization/comms hole opens. Mitigation: drop-when-link-margin-drops policy (drop a node *before* losing the last one), mirroring SubT practice.

- **Downwash in confined spaces** — boid models assume free space; in tight voids, fly looser formations and prefer sequential rather than dense-cluster passage.
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11. Phased Build & Validation Roadmap

1. **Phase 1 — Drift bounding (single drone).** RIO + thermal-inertial + dropped-UWB-anchor localization in a smoke/dust test space. *Success metric:* position error bounded (not growing) over a multi-room loop; altitude error within target.
2. **Phase 2 — Victim detection.** Add perch-and-stare radar vital-signs + thermal/acoustic cueing. *Success metric:* reliable breathing detection through representative debris at 0.5–2.5 m; quantified false-alarm rate.
3. **Phase 3 — Squad + backbone.** 2–3 drones + relay backbone + auction frontier allocation + Class-A alert relay. *Success metric:* victim alert reaches base within seconds across ≥ 2 relay hops after the drone has left direct range.
4. **Phase 4 — Multi-squad + merge.** Squad fracturing at branches + base-station map merge via anchor constraints. *Success metric:* consistent merged map; correctly placed victim markers from independent squads.
5. **Phase 5 — Scale + attrition.** Full fleet, wave-based relaunch, deliberate drone-loss tests. *Success metric:* mission result (victims found, area covered) holds up under the budgeted attrition rate.